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SEAT No. :

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APR -17/ BE/ Insem -23

B.E. (Mechanical/Sandwich) (Semester - II)

COMPUTATIONAL FLUID DYNAMICS (Elective - IV (a))

(2012 Pattern)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer any three questions.*
2) *Figure to the right indicate full marks.*
3) *Neat diagrams must be drawn wherever necessary.*
4) *Use of electronic calculator is allowed.*
5) *Your answer will be valued as a whole.*
6) *Assume suitable data if, necessary.*

Q1) a) State the comments on the governing equations. **[5]**

b) Explain how numbers of equations (of the flow field variables) to the no. of unknowns to solve Navier-Stokes equations are not sufficient. Justify your approach to resolve this. **[5]**

OR

Q2) a) Give examples of automobile and sports application analyses using CFD concepts for application development. **[5]**

b) Mention the significance of substantial derivative in governing equations. **[5]**

Q3) a) Derive finite difference quotient for $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ over a grid having running index (i,j) in X and Y direction respectively, considering first and second order accuracy. **[5]**

b) Given the function $f(x) = 0.25\, x^2$; find the first derivative of f at $x = 2$; using forward, backward and central differencing of order (Δx). Use a step size of $\Delta \times = 0.1$ **[5]**

P.T.O.

OR

Q4) Derive the first derivative of the temperature by finite difference method of a steady heat transfer considering general internal node. **[10]**

Q5) Differentiate two properties of partial differential equations: elliptic, parabolic. **[10]**

OR

Q6) Differentiate grid:

i) Structured vs Unstructured

ii) C and H type grid **[10]**

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